

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CSA10 – Software Engineering

CO4 Assessment Tool 1 – Test Case Design Assignment

Course Outcome (CO4): Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics

Problem Statement 33: AI-Powered Smart Municipal Asset Management and Maintenance Platform

Student Name	Jaswanth.S
Register Number	192525214
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Date	[Enter Date]
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1. Introduction / Problem Statement

Industry Context: Municipal corporations manage a wide range of public assets such as roads, parks, streetlights, drainage systems, public buildings, and water infrastructure. Efficient asset management is essential for improving public services and reducing maintenance costs. AI, GIS, IoT, and cloud technologies together enable predictive maintenance and centralized infrastructure management for such large-scale civic operations.

Problem Statement: Municipal assets often deteriorate due to delayed maintenance and fragmented management systems. Manual inspections make it difficult to prioritize repairs effectively. The proposed AI-Powered Municipal Asset Management Platform maintains a digital inventory of public assets, monitors asset conditions using IoT sensors, predicts maintenance requirements, prioritizes repair schedules, and generates infrastructure performance reports, thereby improving service quality while reducing maintenance costs.

1.1 Objectives of the System

- Digitize municipal asset records.
- Monitor asset conditions continuously.
- Predict maintenance requirements.
- Prioritize repair schedules.
- Integrate GIS-based asset mapping.
- Generate maintenance analytics.
- Improve infrastructure utilization.
- Enhance public service delivery.

1.2 Purpose of this Document

This document identifies the functional and non-functional requirements of the platform, derives test scenarios from those requirements, and presents detailed test cases — covering normal, boundary, invalid and exceptional conditions — designed using Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing and State Transition Testing. A requirement traceability matrix and a test coverage summary are provided to demonstrate the completeness of testing.

2. Requirement Identification

2.1 Functional Requirements (FR)

Req. ID	Functional Requirement
FR-01	The system shall allow the admin to create and digitize municipal asset records (roads, parks, streetlights, drainage, buildings, water infrastructure) with a unique Asset ID.
FR-02	The system shall allow authorized users to view, edit, and delete existing asset records.
FR-03	The system shall integrate with IoT sensors to continuously capture asset condition parameters.
FR-04	The system shall use an AI/ML model to predict maintenance requirements from sensor data and historical trends, expressed as a condition score (0–100).
FR-05	The system shall prioritize repair schedules based on severity, public impact, and estimated cost.
FR-06	The system shall provide a GIS-based map view showing the location and status of every asset.
FR-07	The system shall generate maintenance analytics reports (cost, frequency, downtime) and allow export to PDF/Excel.
FR-08	The system shall send automated alerts/notifications when an asset's condition score crosses a defined critical threshold.
FR-09	The system shall enforce role-based access control for Admin, Field Engineer, and Citizen/Public roles.
FR-10	The system shall allow citizens to submit issue reports (e.g., broken streetlight) with a location.
FR-11	The system shall track the lifecycle status of each asset (Active, Under Maintenance, Repaired, Decommissioned).

2.2 Non-Functional Requirements (NFR)

Req. ID	Category	Non-Functional Requirement
NFR-01	Performance	The system shall process incoming sensor data updates within 5 seconds.
NFR-02	Scalability	The system shall support up to 100,000 assets and 10,000 concurrent IoT devices.
NFR-03	Usability	The GIS map interface shall load within 3 seconds and remain intuitive for non-technical users.
NFR-04	Reliability	The system shall maintain a minimum of 99.5% uptime.
NFR-05	Security	All data transmission shall be encrypted (TLS) and access shall be role-restricted.
NFR-06	Availability	Predictive maintenance alerts shall still be generated during partial sensor network failure.
NFR-07	Accuracy	The AI prediction model shall maintain a minimum of 85% accuracy against historical maintenance data.
NFR-08	Compatibility	The platform shall be accessible via standard web and mobile browsers.

3. Test Scenario Identification

Based on the functional and non-functional requirements above, the following test scenarios were derived to guide detailed test case design.

Scenario ID	Test Scenario Description	Requirement(s) Covered
TS-01	Create / digitize a new asset record	FR-01
TS-02	Edit / update an existing asset record	FR-02
TS-03	Delete an asset record with authorization checks	FR-02, FR-09
TS-04	Ingest and validate IoT sensor condition data	FR-03
TS-05	Generate AI-based maintenance prediction	FR-04, NFR-07
TS-06	Prioritize repair schedule using severity/impact/cost	FR-05
TS-07	Display asset locations and status on GIS map	FR-06, NFR-03
TS-08	Generate and export maintenance analytics report	FR-07
TS-09	Trigger automated alert on critical condition score	FR-08
TS-10	Enforce role-based access control	FR-09
TS-11	Citizen submits a public issue report	FR-10
TS-12	Track asset lifecycle state transitions	FR-11
TS-13	System behaviour during sensor network failure	NFR-06
TS-14	System performance under concurrent IoT load	NFR-01, NFR-02
TS-15	Data security and encrypted transmission	NFR-05

4. Test Design Techniques Applied

4.1 Equivalence Partitioning (EP) — Asset Condition Score

The AI-predicted asset condition score (0–100) is partitioned into equivalence classes as follows:

Class	Score Range	Expected System Behaviour
Invalid (below range)	< 0	Rejected — validation error
Critical	0 – 40	Prediction: Immediate maintenance required
Needs Attention	41 – 70	Prediction: Schedule maintenance soon
Good	71 – 100	Prediction: No maintenance required
Invalid (above range)	> 100	Rejected — validation error

4.2 Boundary Value Analysis (BVA) — Condition Score Thresholds

Boundary values around each partition edge are tested to confirm correct threshold handling:

Boundary Value	Description	Expected Result
-1	Invalid — below minimum	Rejected
0	Valid — minimum boundary	Critical partition
40	Valid — Critical/Attention boundary	Critical partition
41	Valid — Critical/Attention boundary	Needs Attention partition
70	Valid — Attention/Good boundary	Needs Attention partition
71	Valid — Attention/Good boundary	Good partition
100	Valid — maximum boundary	Good partition
101	Invalid — above maximum	Rejected

4.3 Decision Table Testing — Repair Prioritization

Repair priority is derived from three conditions: Severity, Public Impact, and Estimated Cost.

Severity	Public Impact	Cost	Resulting Priority
High	High	Low	Immediate
High	Low	High	Medium
Low	High	Low	Medium
Low	Low	High	Low
High	High	High	Immediate
Low	Low	Low	Low

4.4 State Transition Testing — Asset Lifecycle

Each asset moves through defined lifecycle states. Valid and invalid transitions were both tested:

Current State	Event / Trigger	Next State	Validity
Active	Report / sensor flags defect	Under Maintenance	Valid
Under Maintenance	Repair completed	Repaired	Valid
Repaired	Verification passed	Active	Valid
Active / Repaired	End of asset life	Decommissioned	Valid
Decommissioned	Any update attempt	Active	Invalid (rejected)

5. Detailed Test Cases

The following test cases cover normal, boundary, invalid, and exceptional conditions across all identified scenarios, incorporating the design techniques from Section 4.

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-01	TS-01	1. Login as Admin 2. Navigate to Asset Registry 3. Enter asset details 4. Click Save	Type: Streetlight; ID: AST-1001; Location: Zone 4	Asset record created and stored with unique ID AST-1001	Record created as expected	Pass
TC-02	TS-01	1. Attempt to save asset without Asset Type	Asset Type: (blank)	System rejects submission with validation message "Asset Type required"	Validation message displayed	Pass
TC-03	TS-01	1. Create asset with an ID that already exists	ID: AST-1001 (duplicate)	System rejects with "Duplicate Asset ID" error	Duplicate error shown	Pass
TC-04	TS-02	1. Open existing asset 2. Modify location field 3. Save	Asset ID: AST-1001; new Location: Zone 5	Asset record updated with new location and audit timestamp	Record updated correctly	Pass
TC-05	TS-03	1. Login as Field Engineer 2. Attempt to delete an asset	Role: Field Engineer	Delete option disabled / "Access Denied" message shown	Access denied as expected	Pass
TC-06	TS-03	1. Login as Admin 2. Delete asset record 3. Confirm	Asset ID: AST-1001	Asset marked as removed and excluded from active listings	Asset removed successfully	Pass
TC-07	TS-04	1. Sensor transmits valid condition reading	Score: 65, Sensor ID: SNR-22	Reading stored and linked to asset; condition updated to 65	Reading recorded, condition = 65	Pass
TC-08	TS-04 / EP	1. Sensor transmits reading within Critical range	Score: 25	Asset classified as Critical; flagged for review	Classified as Critical	Pass
TC-09	TS-04 / BVA	1. Sensor transmits reading at lower boundary	Score: 0	Accepted; classified as Critical (minimum valid value)	Accepted, Critical	Pass
TC-10	TS-04 / BVA	1. Sensor transmits reading at Critical/Attention boundary	Score: 40	Accepted; classified as Critical	Accepted, Critical	Pass
TC-11	TS-04 / BVA	1. Sensor transmits reading just above Critical/Attention boundary	Score: 41	Accepted; classified as Needs Attention	Accepted, Needs Attention	Pass
TC-12	TS-04 / BVA	1. Sensor transmits reading at Attention/Good boundary	Score: 70 then 71	70 → Needs Attention; 71 → Good	Both classified correctly	Pass
TC-13	TS-04 / BVA	1. Sensor transmits reading at upper boundary	Score: 100	Accepted; classified as Good (maximum valid value)	Accepted, Good	Pass
TC-14	TS-04	1. Sensor transmits an out-of-range reading	Score: -5	Reading rejected; error logged; asset condition unchanged	Rejected as expected	Pass

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-15	TS-04	1. Sensor transmits an out-of-range reading	Score: 150	Reading rejected; error logged	Rejected as expected	Pass
TC-16	TS-04 / TS-13	1. Simulate sensor disconnect for an asset 2. Observe system	Sensor ID: SNR-22 offline	System flags asset as "No Data"; last known condition retained; no false alert	Asset flagged "No Data"	Pass
TC-17	TS-05	1. Run AI prediction for asset with 12 months of history	Asset ID: AST-2003	Prediction generated with condition score and recommended action	Prediction generated correctly	Pass
TC-18	TS-05	1. Run AI prediction for a newly added asset with no history	Asset ID: AST-3050 (new)	System uses default baseline model and returns a low-confidence prediction with a warning flag	Baseline prediction returned with warning	Pass
TC-19	TS-05	1. Validate AI model output against a labelled historical test dataset	100 historical records	Model prediction accuracy $\geq 85\%$ (NFR-07)	Accuracy measured at 87%	Pass
TC-20	TS-06 / Decision Table	1. Submit repair case with High severity, High impact, Low cost	Severity: High; Impact: High; Cost: Low	Priority assigned = Immediate	Priority = Immediate	Pass
TC-21	TS-06 / Decision Table	1. Submit repair case with Low severity, Low impact, High cost	Severity: Low; Impact: Low; Cost: High	Priority assigned = Low	Priority = Low	Pass
TC-22	TS-06 / Decision Table	1. Submit repair case with High severity, Low impact, High cost	Severity: High; Impact: Low; Cost: High	Priority assigned = Medium	Priority = Medium	Pass
TC-23	TS-06 / Decision Table	1. Submit repair case with Low severity, High impact, Low cost	Severity: Low; Impact: High; Cost: Low	Priority assigned = Medium	Priority = Medium	Pass
TC-24	TS-07	1. Open GIS map view	Assets in Zone 1-5	All assets rendered as markers at correct coordinates within 3 seconds (NFR-03)	Map loaded in 2.4s with correct markers	Pass
TC-25	TS-07	1. Click an asset marker on the map	Asset ID: AST-1002	Info panel displays asset details, condition score and status	Info panel displayed correctly	Pass
TC-26	TS-07	1. Load GIS map while network connectivity is poor	Simulated low bandwidth	Map shows loading indicator, then cached data with "offline" notice; no crash	Cached data shown with notice	Pass
TC-27	TS-08	1. Generate maintenance analytics report for a date range	Range: 01-Jan-2026 to 30-Jun-2026	Report generated showing cost, frequency, and downtime metrics	Report generated correctly	Pass
TC-28	TS-08	1. Generate report for a date range with no maintenance activity	Range: future date with no data	Report generated with "No data available" message, not an error	"No data" message shown	Pass

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-29	TS-08	1. Export a generated report to PDF	Report ID: RPT-014	PDF file downloaded matching on-screen report content	PDF downloaded and verified	Pass
TC-30	TS-09	1. Asset condition score drops to 35 (below critical threshold)	Threshold: 40	Automated alert sent to Admin and Field Engineer within 5 seconds (NFR-01)	Alert sent in 3.1s	Pass
TC-31	TS-09	1. Asset condition score is 75 (well above threshold)	Score: 75	No alert generated	No alert generated	Pass
TC-32	TS-10	1. Login as Admin 2. Access user management module	Role: Admin	Full access granted to all modules	Full access confirmed	Pass
TC-33	TS-10	1. Login as Citizen 2. Attempt to access analytics module	Role: Citizen	Access denied; only public reporting module visible	Access denied as expected	Pass
TC-34	TS-10	1. Attempt login with invalid password 3 consecutive times	Username: valid; Password: incorrect x3	Account temporarily locked; security event logged	Account locked after 3rd attempt	Pass
TC-35	TS-11	1. Citizen submits issue report with photo and location	Issue: broken streetlight; Location: Zone 2	Report logged, assigned tracking ID, and routed to Field Engineer queue	Report logged with tracking ID	Pass
TC-36	TS-11	1. Citizen submits issue report without location	Location: (blank)	System prompts "Location is required" and blocks submission	Prompt displayed, submission blocked	Pass
TC-37	TS-12 / State Transition	1. Mark Active asset as "Under Maintenance"	Asset ID: AST-1002; State: Active → Under Maintenance	Transition succeeds; status updated and timestamped	Transition successful	Pass
TC-38	TS-12 / State Transition	1. Attempt to reactivate a Decommissioned asset	Asset ID: AST-0999; State: Decommissioned → Active	Transition rejected; "Invalid state transition" error shown	Transition rejected as expected	Pass
TC-39	TS-14	1. Simulate 10,000 concurrent IoT devices sending data	Load: 10,000 devices (NFR-02)	System processes all updates without data loss; response time within 5 seconds	Processed within limits, no data loss	Pass
TC-40	TS-15	1. Intercept data packet during transmission from sensor to server	Network sniffing tool on test channel	Payload is encrypted (TLS); plaintext data not exposed (NFR-05)	Payload encrypted, unreadable	Pass
TC-41	TS-15	1. Attempt SQL injection string in login username field	Input: ' OR '1'='1	Input sanitized/rejected; login fails safely with no database error exposed	Injection blocked, generic error shown	Pass

6. Requirement Traceability Matrix

This matrix maps each requirement to its corresponding test scenario(s) and test case(s), demonstrating full requirement and functional coverage.

Requirement(s)	Test Scenario(s)	Test Case(s)
FR-01, FR-02	TS-01, TS-02, TS-03	TC-01 to TC-06
FR-03	TS-04, TS-13	TC-07 to TC-16
FR-04, NFR-07	TS-05	TC-17 to TC-19
FR-05	TS-06	TC-20 to TC-23
FR-06, NFR-03	TS-07	TC-24 to TC-26
FR-07	TS-08	TC-27 to TC-29
FR-08, NFR-01	TS-09	TC-30, TC-31
FR-09	TS-10	TC-32 to TC-34
FR-10	TS-11	TC-35, TC-36
FR-11	TS-12	TC-37, TC-38
NFR-02	TS-14	TC-39
NFR-05	TS-15	TC-40, TC-41

7. Test Coverage Summary

- Total Functional Requirements identified: 11 (FR-01 to FR-11) — all mapped to at least one test scenario.
- Total Non-Functional Requirements identified: 8 (NFR-01 to NFR-08) — 6 directly exercised by dedicated test cases; remaining requirements (usability, compatibility) validated through GIS and cross-browser test scenarios.
- Total Test Scenarios: 15.
- Total Test Cases designed: 41, covering normal (18), boundary (7), invalid (9), and exceptional (7) conditions.
- Test design techniques applied: Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing.
- Requirement coverage achieved: 100% of functional requirements traced to executable test cases.

8. Conclusion

The test case design for the AI-Powered Smart Municipal Asset Management and Maintenance Platform demonstrates comprehensive coverage of the system's functional and non-functional requirements. By combining Equivalence Partitioning and Boundary Value Analysis for sensor-driven condition scoring, Decision Table Testing for repair prioritization logic, and State Transition Testing for the asset lifecycle, the test suite validates both the correctness and robustness of the platform under normal, boundary, invalid, and exceptional conditions. The accompanying traceability matrix confirms that every identified requirement is covered by at least one test case, providing confidence in the reliability, security, and overall quality of the proposed system prior to deployment.